INCH-POUND
MIL-C-19978/21A
18 June 2001
SUPERSEDING
MIL-C-19978/21
30 August 1974

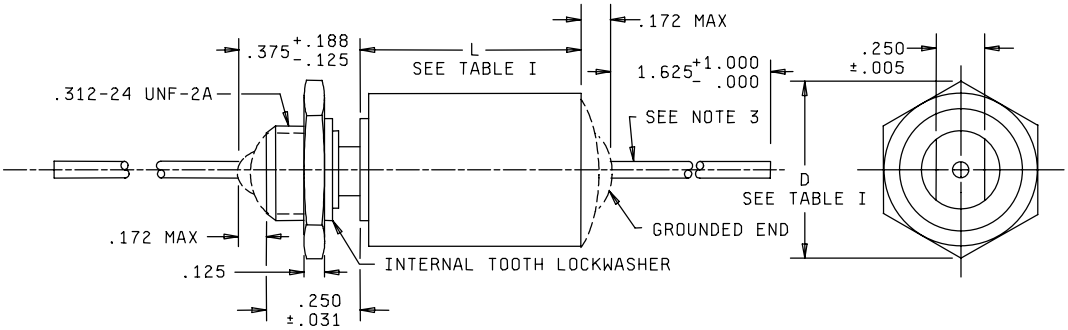
MILITARY SPECIFICATION SHEET

# CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, THREADED-STUD RETAINER, TUBULAR (UNINSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ11

INACTIVE FOR NEW DESIGN
AFTER 30 AUGUST 1974

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-19978.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .004 | 0.10 |
| .005 | 0.13 |
| .031 | 0.79 |
| .032 | 0.81 |
| .125 | 3.18 |
| .172 | 4.37 |
| .250 | 6.35 |
| .312 | 7.92 |
| .375 | 9.53 |
| 1.000 | 25.40 |
| 1.625 | 41.28 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Leads shall be of tinned solid wire, .032 + .004, -.001 (No. 20 AWG).
4. Mounting nuts and lockwashers shall be supplied.

FIGURE 1. Style CQ11 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

TABLE I. Style CQ11 capacitors.

| Part or Identifying Number (PIN) 1/ | DC voltage rating (volts) | Capacitance (µF) | Case dimensions (inches) 2/ | | |
|---|---|---|---|---|---|
| | | | | Circuit 1 | Circuit 3 |
| | | | D+.010 -.020 | L ±.015 | L ±.015 |
| CQ11A-KB333K3 | 100 | .033 | 0.400 | 0.771 | 0.708 |
| CQ11A-KB154K3 | 100 | .15 | 0.400 | 0.896 | 0.833 |
| CQ11A-KB224K3 | 100 | .22 | 0.400 | 1.146 | 1.083 |
| CQ11A-KB394K3 | 100 | .39 | 0.562 | 1.156 | 1.094 |
| CQ11A-KB474K3 | 100 | .47 | 0.562 | 1.156 | 1.094 |
| CQ11A-KB684K3 | 100 | .68 | 0.562 | 1.406 | 1.344 |
| CQ11A-KB105K3 | 100 | 1.0 | 0.562 | 1.906 | 1.844 |
| CQ11A-KC683K3 | 200 | .068 | 0.400 | 0.771 | 0.708 |
| CQ11A-KC104K3 | 200 | .10 | 0.400 | 0.896 | 0.833 |
| CQ11A-KC154K3 | 200 | .15 | 0.400 | 1.146 | 1.083 |
| CQ11A-KC224K3 | 200 | .22 | 0.400 | 1.396 | 1.333 |
| CQ11A-KC334K3 | 200 | .33 | 0.562 | 1.156 | 1.094 |
| CQ11A-KC474K3 | 200 | .47 | 0.562 | 1.406 | 1.344 |
| CQ11A-KC684K3 | 200 | .68 | 0.562 | 1.906 | 1.844 |
| CQ11A-KC105K3 | 200 | 1.0 | 0.670 | 1.906 | 1.844 |
| CQ11A-KE103K3 | 400 | .010 | 0.400 | 0.771 | 0.708 |
| CQ11A-KE393K3 | 400 | .039 | 0.400 | 0.896 | 0.833 |
| CQ11A-KE473K3 | 400 | .047 | 0.400 | 0.896 | 0.833 |
| CQ11A-KE683K3 | 400 | .068 | 0.400 | 1.146 | 1.083 |
| CQ11A-KE104K3 | 400 | .10 | 0.400 | 1.396 | 1.333 |
| CQ11A-KE154K3 | 400 | .15 | 0.562 | 1.156 | 1.094 |
| CQ11A-KE224K3 | 400 | .22 | 0.562 | 1.406 | 1.344 |
| CQ11A-KE274K3 | 400 | .27 | 0.562 | 1.906 | 1.844 |
| CQ11A-KE334K3 | 400 | .33 | 0.562 | 1.906 | 1.844 |
| CQ11A-KE474K3 | 400 | .47 | 0.670 | 1.906 | 1.844 |
| CQ11A-KE684K3 | 400 | .68 | 0.750 | 2.156 | 2.094 |
| CQ11A-KE105K3 | 400 | 1.0 | 1.000 | 1.906 | 1.844 |
| CQ11A-KF102K3 | 600 | .001 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF152K3 | 600 | .0015 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF222K3 | 600 | .0022 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF332K3 | 600 | .0033 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF472K3 | 600 | .0047 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF682K3 | 600 | .0068 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF103K3 | 600 | .010 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF153K3 | 600 | .015 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF223K3 | 600 | .022 | 0.400 | 0.771 | 0.708 |
| CQ11A-KF333K3 | 600 | .033 | 0.400 | 0.896 | 0.833 |
| CQ11A-KF473K3 | 600 | .047 | 0.400 | 1.146 | 1.083 |
| CQ11A-KF683K3 | 600 | .068 | 0.400 | 1.396 | 1.333 |
| CQ11A-KF104K3 | 600 | .10 | 0.562 | 1.156 | 1.094 |
| CQ11A-KF154K3 | 600 | .15 | 0.562 | 1.406 | 1.344 |
| CQ11A-KF184K3 | 600 | .18 | 0.562 | 1.906 | 1.844 |
| CQ11A-KF224K3 | 600 | .22 | 0.562 | 1.906 | 1.844 |
| CQ11A-KF334K3 | 600 | .33 | 0.670 | 1.906 | 1.844 |
| CQ11A-KF474K3 | 600 | .47 | 0.750 | 2.156 | 2.094 |
| CQ11A-KF684K3 | 600 | .68 | 1.000 | 1.906 | 1.844 |

See footnotes on next page.



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TABLE I. Style CQ11 capacitors - Continued.

1/ Complete part number shall include an additional symbol to indicate circuit.
2/ See table II for metric equivalents.

TABLE II. Millimeter equivalent of decimal inches.

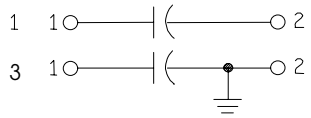
| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| 0.010 | 0.25 | 0.670 | 17.02 | 0.896 | 22.76 | 1.156 | 29.36 | 1.844 | 46.84 |
| 0.015 | 0.38 | 0.708 | 17.98 | 1.000 | 25.40 | 1.333 | 33.86 | 1.906 | 48.41 |
| 0.020 | 0.51 | 0.750 | 19.05 | 1.083 | 27.51 | 1.344 | 34.14 | 2.094 | 53.19 |
| 0.400 | 10.16 | 0.771 | 19.58 | 1.094 | 27.79 | 1.396 | 35.46 | 2.156 | 54.76 |
| 0.562 | 14.27 | 0.833 | 21.16 | 1.146 | 29.11 | 1.406 | 35.71 | | |

Case material: Nonmagnetic (screwneck and end seal may be of magnetic material).

Dielectric material: Paper and polyethelene terephthalate (K).

Terminals: Axial-wire lead (see figure 1).

Circuit diagram:




Rated voltage: See table I.

Rated temperature: -65° to +125°C.

Capacitance (nominal): See table I.

Capacitance tolerance: ±10 percent (symbol K).

Vibration grade: 3.

Dissipation factor (DF) (max): 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter A.

Barometric pressure, qualification only: In accordance with MIL-PRF-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Circuit diagram 3: Between ungrounded terminal and case.

Insulation resistance (IR):

Terminal to terminal: See table III.



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TABLE III. Terminal-to-terminal insulation resistance.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| | At 25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At 125°C |
| 0 to 0.08 microfarad | 250 megohms |
| Greater than 0.08 microfarad | 20 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminal to case: Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:

Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-PRF-19978.

Solderability: In accordance with MIL-PRF-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter D.

Stability at low and high temperature:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

High temperature:

Test temperature: +125°C +5°C, -0°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-PRF-19978.

NOTE: Style CQ11 capacitors are substitutable for style CP11 capacitors formerly covered in MIL-C-25/1K as shown in the following:



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| From MIL-C-25/1K (CP11) | To MIL-C-19978/21 (CQ11) | From MIL-C-25/1K (CP11) | To MIL-C-19978/21 (CQ11) |
|---|---|---|---|
| CP11A-KB333K- | CQ11A-KB333K3- | CP11A-KE334K- | CQ11A-KE334K3- |
| CP11A-KB154K- | CQ11A-KB154K3- | CP11A-KE474K- | CQ11A-KE474K3- |
| CP11A-KB224K- | CQ11A-KB224K3- | CP11A-KE684K- | CQ11A-KE684K3- |
| CP11A-KB394K- | CQ11A-KB394K3- | CP11A-KE105K- | CQ11A-KE105K3- |
| CP11A-KB474K- | CQ11A-KB474K3- | CP11A-KF102K- | CQ11A-KF102K3- |
| CP11A-KB684K- | CQ11A-KB684K3- | CP11A-KF152K- | CQ11A-KF152K3- |
| CP11A-KB105K- | CQ11A-KB105K3- | CP11A-KF222K- | CQ11A-KF222K3- |
| CP11A-KC683K- | CQ11A-KC683K3- | CP11A-KF332K- | CQ11A-KF332K3- |
| CP11A-KC104K- | CQ11A-KC104K3- | CP11A-KF472K- | CQ11A-KF472K3- |
| CP11A-KC154K- | CQ11A-KC154K3- | CP11A-KF682K- | CQ11A-KF682K3- |
| CP11A-KC224K- | CQ11A-KC224K3- | CP11A-KF103K- | CQ11A-KF103K3- |
| CP11A-KC334K- | CQ11A-KC334K3- | CP11A-KF153K- | CQ11A-KF153K3- |
| CP11A-KC474K- | CQ11A-KC474K3- | CP11A-KF223K- | CQ11A-KF223K3- |
| CP11A-KC684K- | CQ11A-KC684K3- | CP11A-KF333K- | CQ11A-KF333K3- |
| CP11A-KC105K- | CQ11A-KC105K3- | CP11A-KF473K- | CQ11A-KF473K3- |
| CP11A-KE103K- | CQ11A-KE103K3- | CP11A-KF683K- | CQ11A-KF683K3- |
| CP11A-KE393K- | CQ11A-KE393K3- | CP11A-KF104K- | CQ11A-KF104K3- |
| CP11A-KE473K- | CQ11A-KE473K3- | CP11A-KF154K- | CQ11A-KF154K3- |
| CP11A-KE683K- | CQ11A-KE683K3- | CP11A-KF184K- | CQ11A-KF184K3- |
| CP11A-KE104K- | CQ11A-KE104K3- | CP11A-KF224K- | CQ11A-KF224K3- |
| CP11A-KE154K- | CQ11A-KE154K3- | CP11A-KF334K- | CQ11A-KF334K3- |
| CP11A-KE224K- | CQ11A-KE224K3- | CP11A-KF474K- | CQ11A-KF474K3- |
| CP11A-KE274K- | CQ11A-KE274K3- | CP11A-KF684K- | CQ11A-KF684K3- |

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Navy - AS

Preparing activity:
DLA - CC

(Project 5910-2118-07)



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